

PARTH BALAR

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EXPERIENCE

Site Reliability Engineer (Contract) | Splunk · Ahmedabad, India

Jun 2022 – Dec 2023

- Owned end-to-end SLO/error-budget program for Splunk Cloud; maintained 99.9% uptime via proactive alerting and automated Python/Shell remediation across distributed systems at scale.
- Built GitOps CI/CD pipelines with Jenkins and Terraform enabling zero-downtime blue-green releases on Kubernetes (AWS & GCP), cutting deployment cycle time by 40%.
- Authored reusable Terraform IaC modules that reduced environment provisioning from days to hours (60% faster), adopted as team standard for all new service onboarding.
- Rearchitected indexing/ingestion pipelines, reducing query latency by 30% and ingestion overhead by 20%; built Grafana/Splunk dashboards surfacing degradation before user impact.
- Led blameless postmortems and chaos-engineering reviews; converted learnings into runbooks and automated recovery playbooks, cutting MTTR by 40% over six months.
- Drove FinOps initiative (AWS/GCP migration + rightsizing) via capacity modeling, achieving 15% reduction in cloud spend.

Site Reliability Engineer Intern | Crest Data · Ahmedabad, India

Dec 2021 – May 2022

- Containerized legacy services with Docker and deployed on Kubernetes (AWS); automated provisioning via Terraform + Jenkins, reducing deployment setup from hours to minutes.
- Built an internal automation platform (FastAPI + React) replacing manual data-processing workflows, delivering 60% reduction in manual effort and 40% fewer data errors.
- Introduced structured observability with Grafana and Splunk (SLIs/SLO targets + postmortem culture), reducing MTTR by 35% within the first quarter.

PROJECTS

FRIDAY - Autonomous AI Assistant with Full Computer Control · Python · FastAPI · LangGraph · LangChain · React · ChromaDB · Docker · WebSocket

- Architected a full-stack AI assistant (Python 3.13 + FastAPI backend, React 18 frontend) powered by Claude Sonnet 4.6, exposing 112+ LangChain tools across 36 specialized modules.
- Designed a LangGraph ReAct autonomous agent with multi-step planning; built a two-tier intent router (regex <1 ms + LLM fallback) handling 55+ intents across email, file I/O, browser automation, and system control.
- Implemented semantic memory with ChromaDB, a Markov-chain predictive prefetch engine, and a nightly self-reflection loop (Claude Haiku) that evolves the system prompt based on user correction patterns.
- Engineered a 5-source parallel job-search pipeline (ThreadPoolExecutor) with yfinance company-health scoring and SQLite-backed application tracking; deployed via Docker Compose with Nginx reverse proxy.

Graph-Enhanced RAG Retrieval System · Python · sentence-transformers · ChromaDB · Knowledge Graph · BFS · Cosine Similarity

- Built a hybrid retrieval pipeline combining dense vector search (all-MiniLM-L6-v2 embeddings, cosine similarity) with BFS knowledge-graph traversal to surface structurally related code chunks invisible to keyword search.
- Pre-computed L2-normalised entity embeddings enabling single matrix-multiply similarity scoring across 80+ entities; merged graph traversal chunks with standard vector results for up to 50 deduplicated hits per query.

DataZen — Self-Serve Analytics Platform · Python · React · ML · Data Visualization

- Designed a platform transforming raw CSV uploads into business insights in under 1 minute with auto-generated visualizations, RFM customer segmentation, real-time inventory alerts, and AI-generated executive summaries.

EDUCATION

M.S. Information Technology Management · The University of Texas at Dallas

Aug 2023 – May 2025

B.S. Information Technology · Dharmsinh Desai University, India

Jun 2018 – May 2022

TECHNICAL SKILLS

Languages: Python, Shell Scripting, JavaScript, SQL, Java, C/C++

Cloud & Infrastructure: AWS (primary), GCP · Kubernetes, Docker, ECS · Terraform, Ansible

SRE & Reliability: SLO/SLI/Error Budget, Chaos Engineering, Incident Management, Blameless Postmortems

Observability: Splunk (Cloud & Enterprise), Grafana, Prometheus, ELK Stack, CloudWatch

CI/CD & GitOps: Jenkins, GitHub Actions, GitLab CI/CD · Blue-Green & Canary Deployments

AI/ML Frameworks: LangGraph, LangChain, LangChain Anthropic, ChromaDB, sentence-transformers

Databases: PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB (vector)

Web & APIs: FastAPI, Flask, Django, React.js, Node.js